1. A Novel Data-Centric Programming Model for Large-Scale Parallel Systems

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Abstract: This paper presents the main features and the programming constructs of the DCEx programming model designed for the implementation of data-centric large-scale parallel applications on Exascale computing platforms. To support scalable parallelism, the DCEx programming model employs private data structures and limits the amount of shared data among parallel threads. The basic idea of DCEx is structuring programs into data-parallel blocks to be managed by a large number of parallel threads. Parallel blocks are the units of shared- and distributed-memory parallel computation, communication, and migration in the memory/storage hierarchy. Threads execute close to data using near-data synchronization according to the PGAS model. A use case is also discussed showing the DCEx features for Exascale programming. © 2020, Springer Nature Switzerland AG.
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